IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

• **The following methodologies were used to analyze data:**
•Collection of structured data through **web scraping** and the **SpaceX API**, allowing the creation of a comprehensive launch dataset.
• **Data Collection using web scraping and SpaceX API:**
•Information on mission outcomes, launch pads, rocket types, and landing attempts was obtained from publicly available sources.
• **Exploratory Data Analysis (EDA), including data wrangling, data visualization and interactive visual analytics:**
•Data was cleaned and transformed to ensure quality.
•Visual and interactive tools were used to detect trends and correlations.
• **Machine Learning Prediction:**
•Several ML models were tested to predict **launch success**.
•The models identified the most relevant features impacting the outcome.
• **Summary of all results:**
•Public data was valuable for building predictive models.
•EDA helped isolate the key factors for successful missions.
•ML revealed the optimal strategy to forecast and improve launch success.

Introduction

• The objective is to evaluate the viability of the new company Space Y to compete with SpaceX:

•The analysis aims to determine whether **Space Y** could become a competitive player in the commercial space launch industry by leveraging data-driven insights.

• Desirable answers:

•**What is the most accurate way to estimate total launch costs**, particularly by predicting the likelihood of successful landings of the rocket's first stage — a key factor in cost reduction and mission reusability.

•**Which launch sites offer the best conditions** for mission success, helping to optimize operational efficiency and strategic planning.

Section 1

Methodology

Methodology

Executive Summary

•**Data Collection Methodology:**

Data on SpaceX launches was obtained from two main sources:

• SpaceX API (https://api.spacexdata.com/v4/rockets/)

• Web scraping from Wikipedia (https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)

•**Data Wrangling:**

The dataset was cleaned and enhanced by creating a **landing outcome label**, derived from outcome data after summarizing and analyzing key features.

•**Exploratory Data Analysis (EDA):**

Conducted using **visualization techniques** and **SQL queries** to identify relevant patterns and relationships.

Methodology

•**Interactive Visual Analytics:**
Tools like **Folium** and **Plotly Dash** were used to develop interactive visualizations for deeper insights.
•**Predictive Analysis:**
The data was **normalized** and split into **training and testing sets**.
Four **classification models** were tested, with accuracy evaluated using different parameter settings todetermine the most effective approach.

Data Collection

Launch data was gathered from two primary sources: the **SpaceX API** (https://api.spacexdata.com/v4/rockets/) and **Wikipedia** (https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches).

Information was extracted using **web scraping methods**, allowing the compilation of a rich dataset covering rocket specifications, missions, and outcomes.

Data Collection – SpaceX API

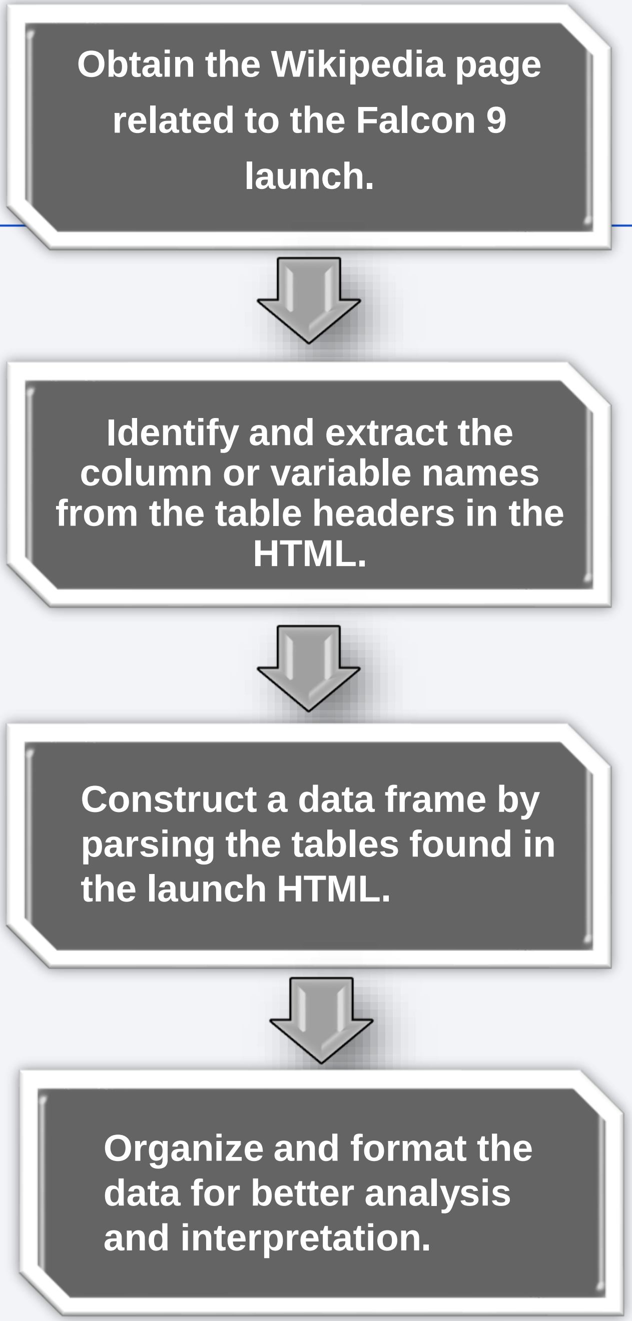
•**SpaceX provides a public API** that allows access to detailed data on rockets, launches, and landing attempts. This data can be retrieved programmatically and used for analysis.
•The **data extraction process followed a specific flowchart**, guiding the steps from API connection to data transformation and storage for further use.
•Once collected, the data was **persisted locally** to enable repeated access and facilitate cleaning, analysis, and modeling without needing constant API calls.

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/jupyter-labs-spacex-data-collection-api%20(1).ipynb

Data Collection - Scraping

•**Data on SpaceX launches** can also be sourced from **Wikipedia**, offering a complementary set of information on individual missions and launch history.
•The **data is downloaded** from Wikipedia following the process outlined in the flowchart, after which it is **persisted** for further analysis.
•After persistence, the data was **cleaned and enriched** to ensure consistency and improve its quality for modeling and analysis.

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/jupyter-labs-webscraping.ipynb



Data Wrangling

•**Exploratory Data Analysis (EDA)** was initially conducted on the dataset to uncover patterns and insights.
•**Summaries were then generated**, including the number of launches per site, the frequency of each orbit type, and the distribution of mission outcomes by orbit type.
•Finally, a **landing outcome label** was created based on the values from the **Outcome** column to categorize the success or failure of each landing attempt.

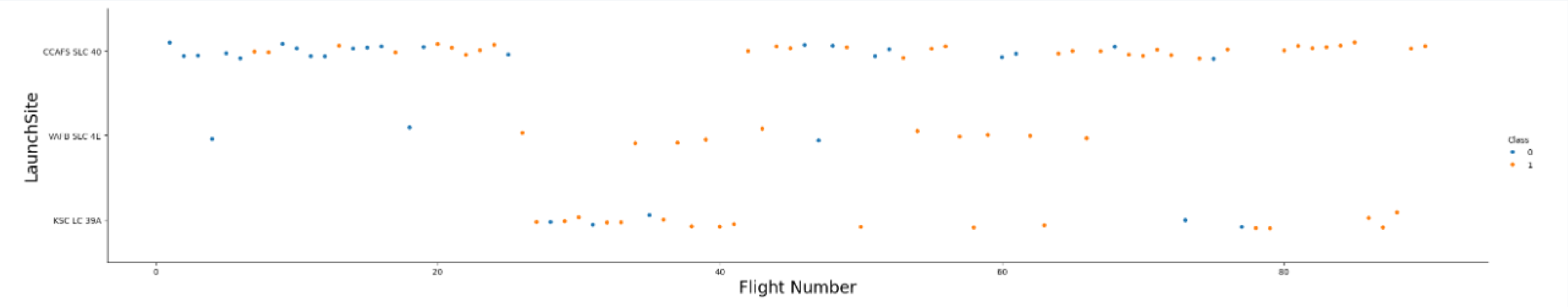
https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

Scatterplots and barplots were used to explore the data and visualize the relationships between pairs of features, such as:

- **Payload Mass vs. Flight Number**
- **Launch Site vs. Flight Number**
- **Launch Site vs. Payload Mass**
- **Orbit vs. Flight Number**
- **Payload vs. Orbit**

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/edadataviz.ipynb



EDA with SQL

The following **SQL queries** were executed to extract key insights from the dataset:

- **Names of unique launch sites** involved in space missions.
- **Top 5 launch sites** whose names begin with "CCA".
- **Total payload mass** carried by boosters launched by NASA (CRS).
- **Average payload mass** carried by the booster version F9 v1.1.
- **Date of the first successful landing** outcome on a ground pad.
- **Booster names** that achieved success on a drone ship with payload mass between 4000 and 6000 kg.
- **Total count of successful and failed mission outcomes**.
- **Booster versions** that carried the maximum payload mass.
- **Failed landing outcomes** on the drone ship in 2015, including booster versions and launch site names.
- **Ranking of landing outcomes** (e.g., Failure on drone ship or Success on ground pad) between **2010-06-04 and 2017-03-20**.

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Folium Maps was used to visualize spatial data through various map elements:

- **Markers** were placed to indicate key points, such as **launch sites**.
- **Circles** were used to highlight **specific areas** around coordinates, like the **NASA Johnson Space Center**.
- **Marker clusters** grouped **events** at the same coordinates, such as multiple **launches** occurring at a single **launch site**.
- **Lines** were drawn to represent **distances** between two coordinates, providing spatial context for launches.

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

To better understand the data, several **visualizations** were created:

• A **bar chart showing the percentage of launches by site** to highlight the most frequently used locations.

• A **distribution plot of payload ranges** to analyze the variation in payload masses.

•By combining these visuals, it was possible to **identify patterns between payload sizes and launch sites**, providing insight into which locations are most suitable for different types of missions.

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/spacex_dashboard.py

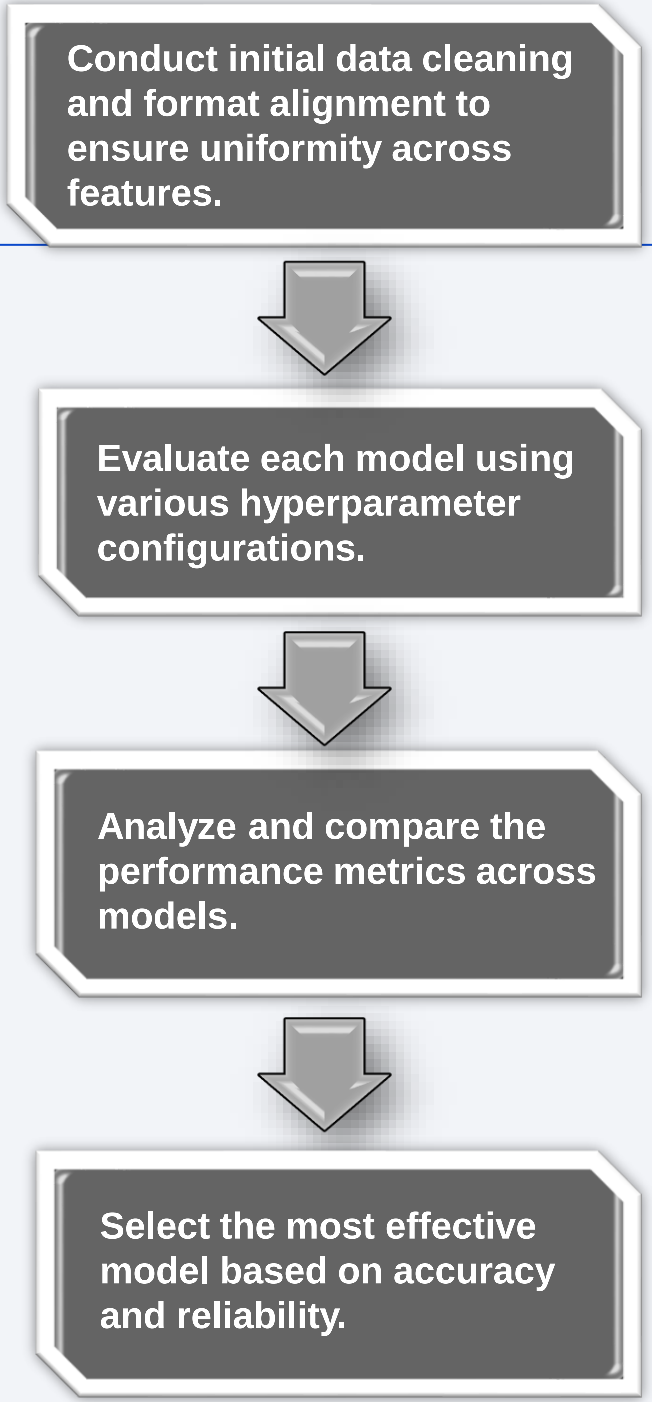
Predictive Analysis (Classification)

A comparison was made between **four classification models** to evaluate their performance in predicting landing success:

- **Logistic Regression**
- **Support Vector Machine (SVM)**
- **Decision Tree**
- **K-Nearest Neighbors (KNN)**

Each model was assessed based on its **accuracy and effectiveness** in identifying the key factors influencing successful landings.

https://github.com/chrismedina98/IBM-Applied-Data-Science-Capstone/blob/d95de6d405d720e122a4286d1787699be1b44180/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb



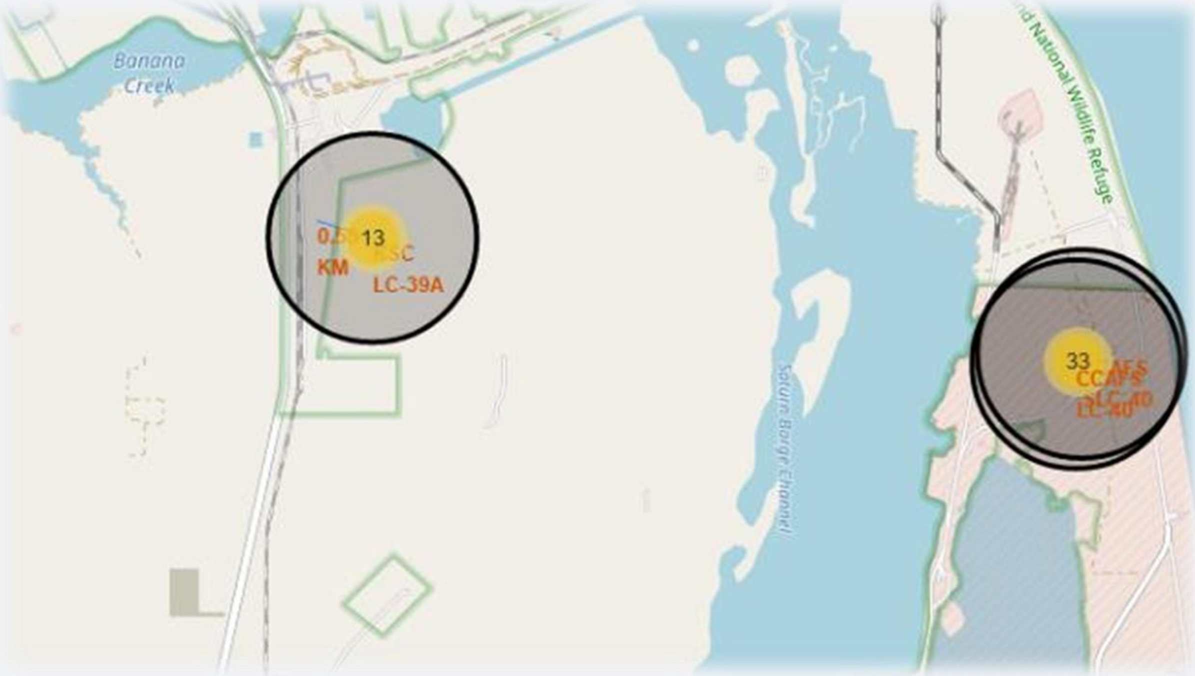
Results

Exploratory Data Analysis – Key Findings:

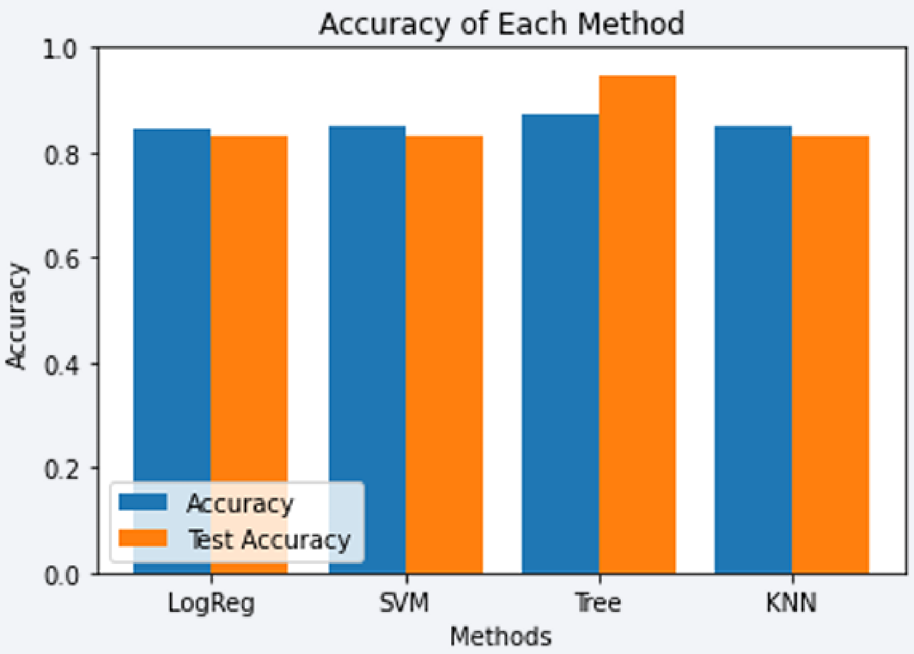
•**SpaceX operates from four distinct launch sites**, each contributing to its mission portfolio.
•The **initial launches** were carried out primarily for **SpaceX** and **NASA** missions.
•The **average payload mass** for the **F9 v1.1 booster** is approximately **2,928 kg**.
•The **first successful landing** occurred in **2015**, five years after SpaceX's first recorded launch.
•**Several Falcon 9 booster versions** successfully landed on **drone ships**, often carrying payloads **above the average**.
•The company achieved **an almost perfect mission success rate**, nearing **100%**.
•In **2015**, two boosters—**F9 v1.1 B1012** and **F9 v1.1 B1015—failed to land** on drone ships.
•**Landing outcomes improved progressively** over time, showing clear advances in reliability and technology.

Results


United States
Washington
10
Los Angeles
SLC-4E
Phoenix
46
AFS
The Bahamas
México
La Habana
Cuba
Ciudad
de México
República
Dominicana
Kingston
Ciudad
de Guatemala
Honduras
Nicaragua


Banana
Creek
13
KM
LC-39A
33
CCAFS
Saturn Barge Channel

Results



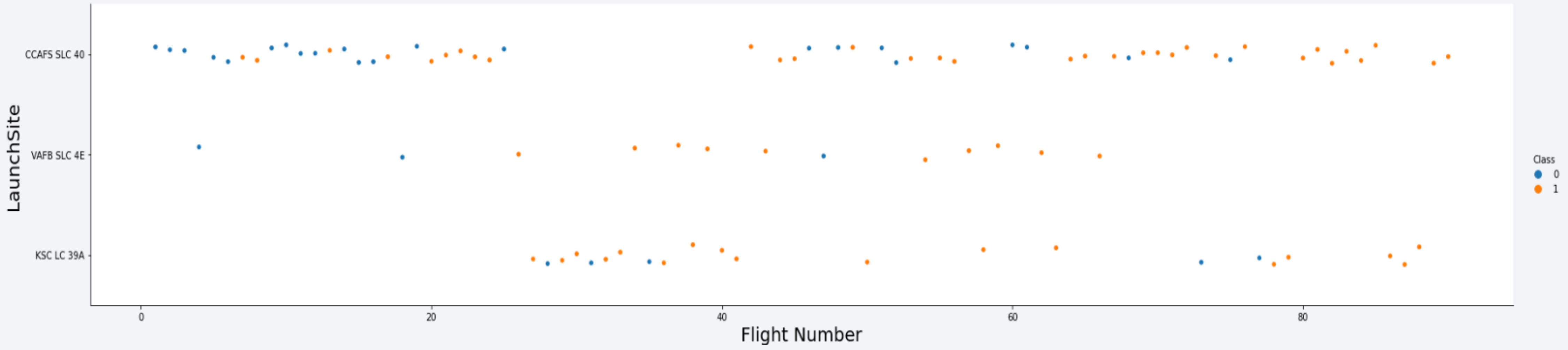
•**Predictive analysis revealed** that the **Decision Tree Classifier** was the most effective model for forecasting successful landings.
•It achieved an **accuracy of over 87% during training** and **over 94% on test data**, demonstrating strong generalization and predictive performance.

Section 2

Insights drawn from EDA

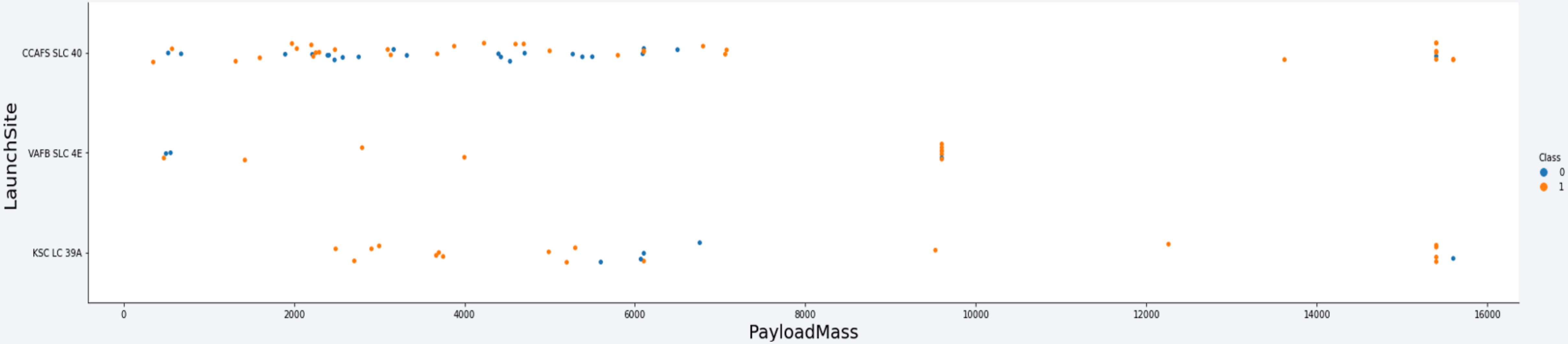
Flight Number vs. Launch Site

•The plot indicates that **CCAFS SLC-40** is currently the **most successful launch site**, hosting the majority of recent successful missions.
•**VAFB SLC-4E** ranks second, followed by **KSC LC-39A** in third place.
•The visualization also shows a **clear upward trend in success rates over time**, reflecting SpaceX's continuous improvements in launch reliability.



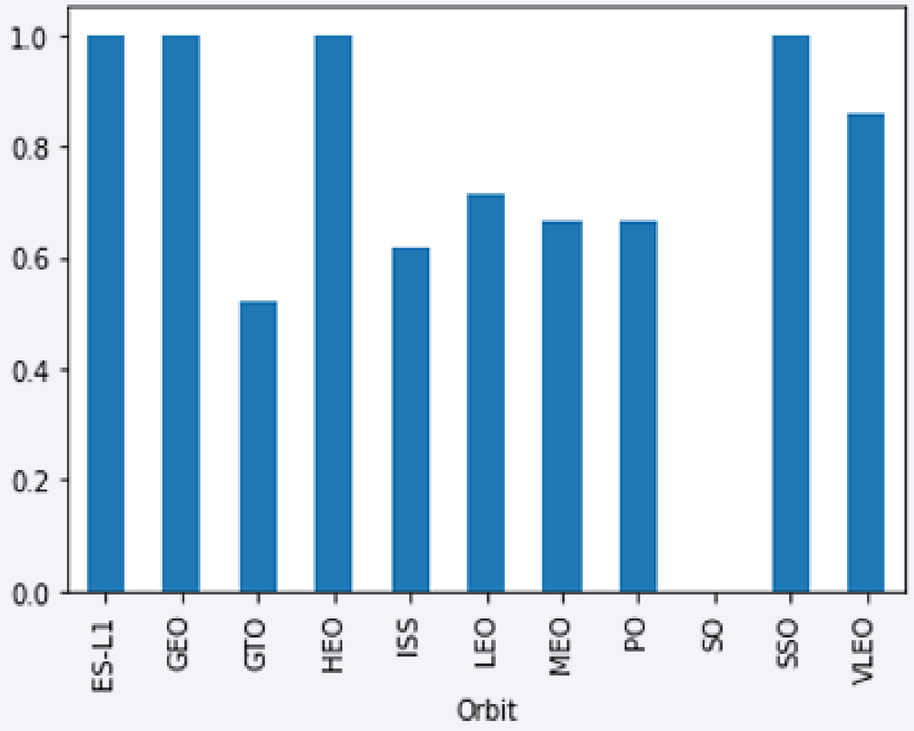
Payload vs. Launch Site

•**Payloads exceeding 9,000 kg** (roughly the weight of a school bus) show a **very high success rate**.
•**Payloads above 12,000 kg** appear to be **launched exclusively** from **CCAFS SLC-40** and **KSC LC-39A**, likely due to their advanced infrastructure and capacity.



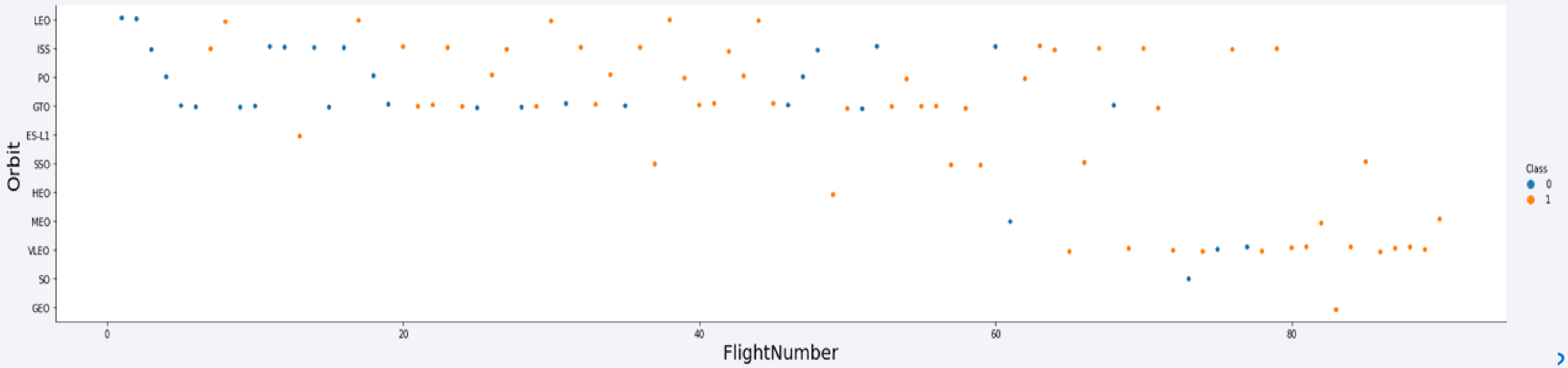
Success Rate vs. Orbit Type

•The **highest success rates** were observed in missions targeting the following orbits:
• **ES-L1 100%**
• **GEO 100%**
• **HEO 100%**
• **SSO 100%**
• **VLEO 80%**
• **LFO 70%**



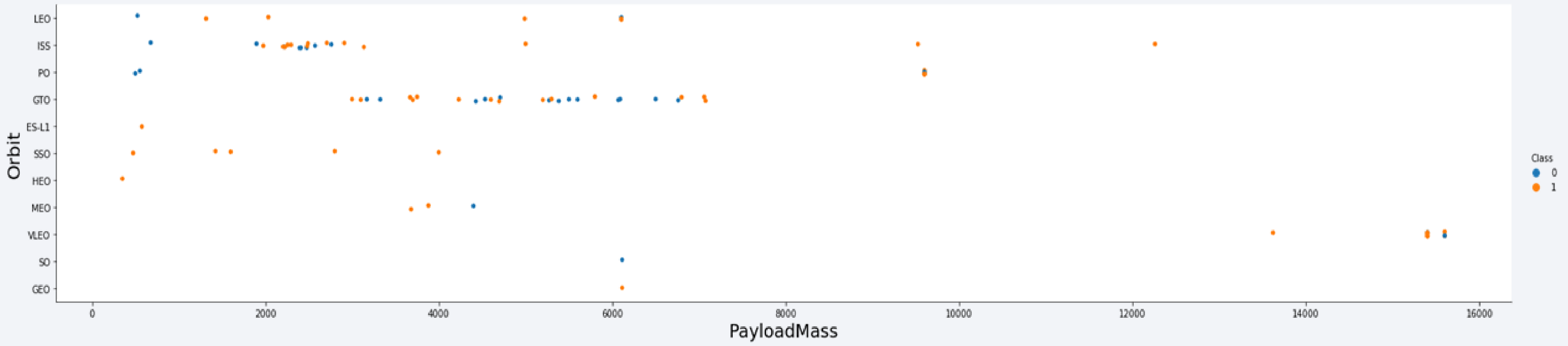
Flight Number vs. Orbit Type

•The **growing number of missions to VLEO orbit** suggests a **rising interest and potential market opportunity**, possibly driven by applications like satellite constellations for communication and Earth observation.
•**Increased mission frequency to specific orbits** like VLEO and SSO may indicate **strategic shifts** in commercial and scientific priorities.
•The data also hints that **SpaceX's reliability improvements** are enabling the safe transport of heavier payloads to more diverse orbits.



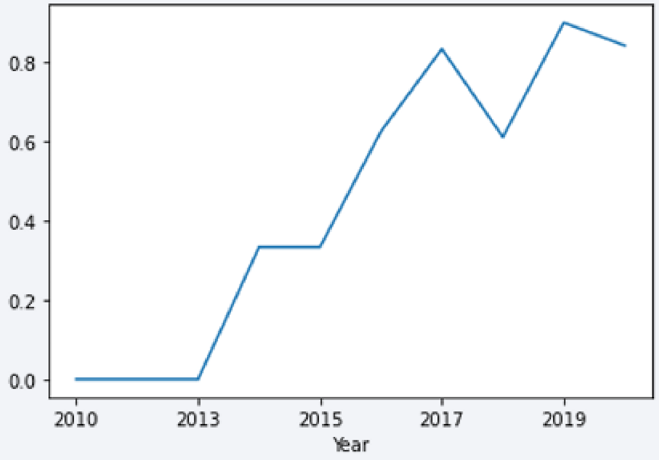
Payload vs. Orbit Type

- There appears to be **no clear correlation between payload mass and success rate** for missions targeting the **GTO (Geostationary Transfer Orbit)**.
- The **ISS orbit** shows a **broad range of payloads** being delivered, maintaining a **strong success rate**, which reflects its importance and reliability in mission planning.
- **SO and GEO orbits** have had **relatively few launches**, suggesting they are either **less in demand** or **more specialized** in nature.
- The diversity in payloads to ISS may indicate **operational flexibility**, while the low number of missions to SO and GEO could highlight **technical challenges or limited commercial use**.



Launch Success Yearly Trend

•**Success rates began to rise notably from 2013 onward**, showing a steady improvement through **2020**.
•The **initial three years** appear to have served as a **phase of adaptation and technological refinement**, laying the foundation for more consistent mission success in later years.



All Launch Site Names

•Based on the data, there are **four distinct launch sites**:
• **CCAFS LC-40**
• **CCAFS SLC-40**
• **KSC LC-39A**
• **VAFB SLC-4E**
•These launch sites were identified by selecting the **unique values** in the **"launch_site"** column of the dataset.

Launch Site Names Begin with 'CCA'

Five records were identified where the **launch sites** begin with the string **"CCA“:**

Date	Time UTC	Booster Version	Launch Site	Payload	Payload Mass kg	Orbit	Customer	Mission Outcome	Landing Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attemp

Total Payload Mass

Total Payload Carried by NASA Boosters:
The total payload was calculated by summing the payloads associated with booster codes containing **‘CRS’**, which corresponds to NASA missions.
Total Payload (kg): 111.268

Total Payload (kg)
111.268

Average Payload Mass by F9 v1.1

Average Payload Mass Carried by Booster Version F9 v1.1:
By filtering the data based on the booster version and calculating the average payload mass, we obtained an average value of **2,928 kg** for the F9 v1.1 booster.

Avg Payload (kg)
2.928

First Successful Ground Landing Date

First Successful Landing Outcome on Ground Pad:
By filtering the data for successful landing outcomes on the ground pad and selecting the minimum date, we identified that the first successful landing occurred on **12/22/2015**.

Min Date
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Boosters Successfully Landed on Drone Ship with Payload Mass Between 4,000 kg and 6,000 kg:
By selecting distinct booster versions based on the filters above, the following 4 boosters meet the criteria of successfully landing on a drone ship and carrying a payload mass greater than 4,000 kg but less than 6,000 kg.

Booster Version
F9 FT B1021.2
F9 FT B1031.2
F9 FT B1022
F9 FT B1026

Total Number of Successful and Failure Mission Outcomes

Number of Successful and Failed Mission Outcomes:
By grouping the mission outcomes and counting the records for each group, we obtained the summary of successful and failed mission outcomes as shown above.

Mission Outcome	Occurrences
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

Boosters Carried Maximum Payload

Boosters that Have Carried the Maximum Payload Mass:
These are the boosters that have carried the highest payload mass recorded in the dataset.

Booster Version (...)
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3

Booster Version
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

2015 Launch Records

Failed Landing Outcomes on Drone Ship in 2015:
The list shows the only two occurrences of failed landing outcomes on a drone ship, along with their corresponding booster versions and launch site names for the year 2015.

Booster Version	Launch Site
F9 v1.1 B1012	CCAFS LC-40
F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Ranking of All Landing Outcomes Between 2010-06-04 and 2017-03-20:
This data view highlights that the "No attempt" category must also be taken into account when analyzing landing outcomes during this period.

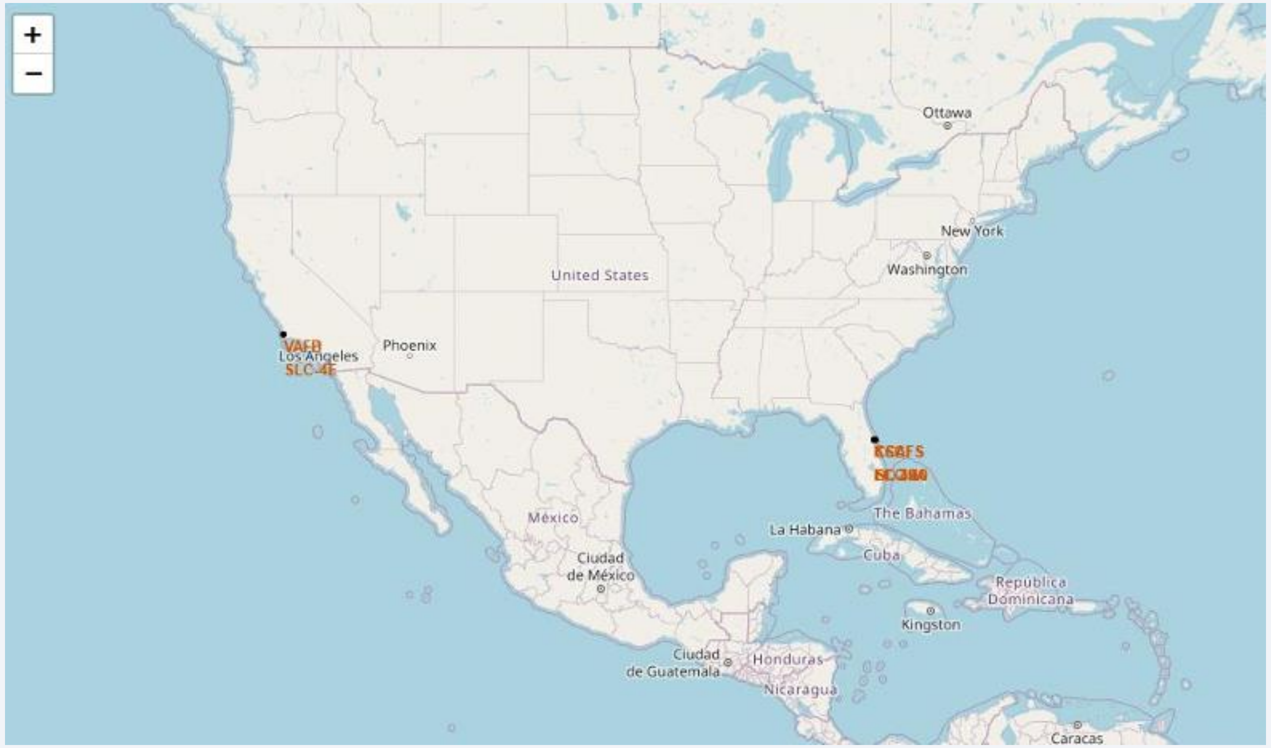
Landing Outcome	Occurrences
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Launch Sites

•**Launch sites are typically located near the coast**, likely for **safety reasons**, allowing rockets to launch over open water.
•However, they are **strategically positioned not too far from roads and railroads**, ensuring **easy access for logistics, transportation**, and support infrastructure.

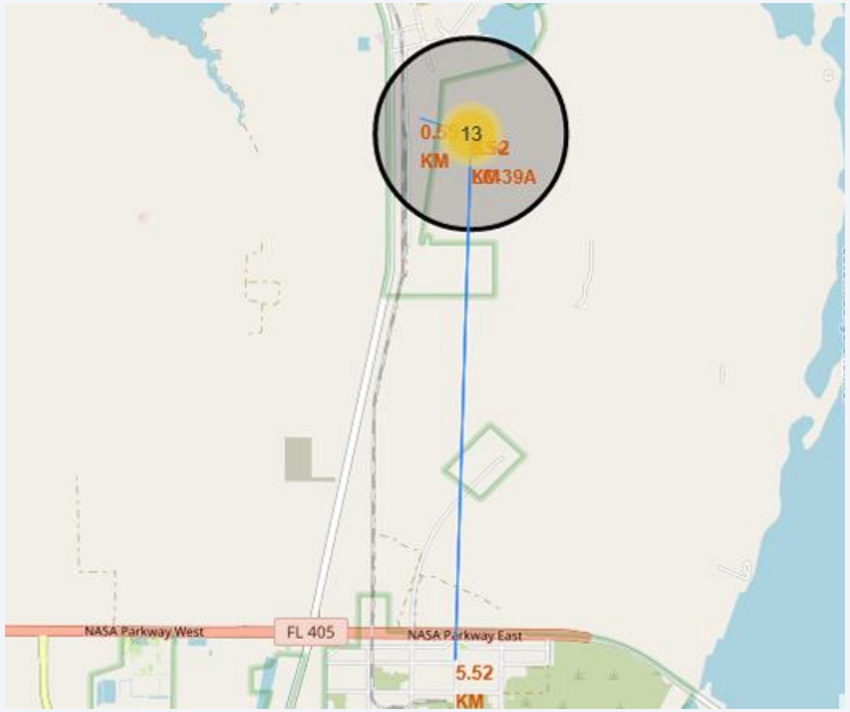


Launch Outcomes

•An example of the **KSC LC-39A launch site** shows a variety of **launch outcomes**, including both successful and unsuccessful missions.
•The **green points** represent **successful launches**, while the **red points** indicate **failed missions**.
•This site has been used for **numerous high-profile launches**, highlighting its key role in SpaceX's operational strategy.

Logistic and Safety

•The **KSC LC-39A launch site** offers **strong logistical advantages**, as it is **near roads and railroads**, providing easy access for transportation and support.
•Additionally, it is **relatively distant from populated areas**, ensuring **safety** for both personnel and surrounding communities during launches.

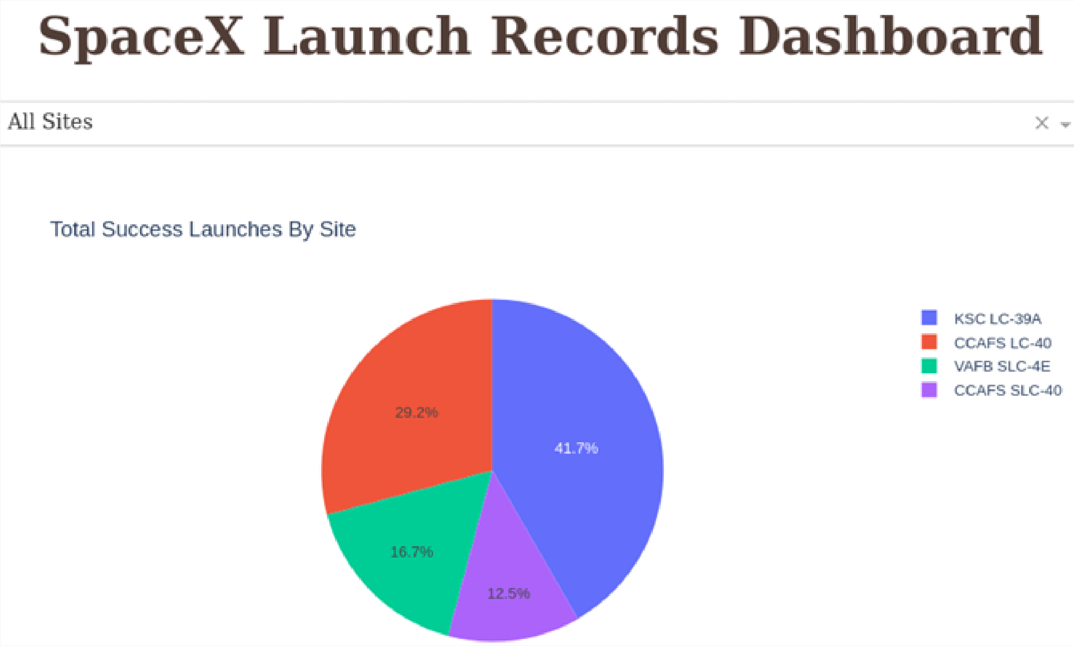


Section 4

Build a Dashboard with Plotly Dash

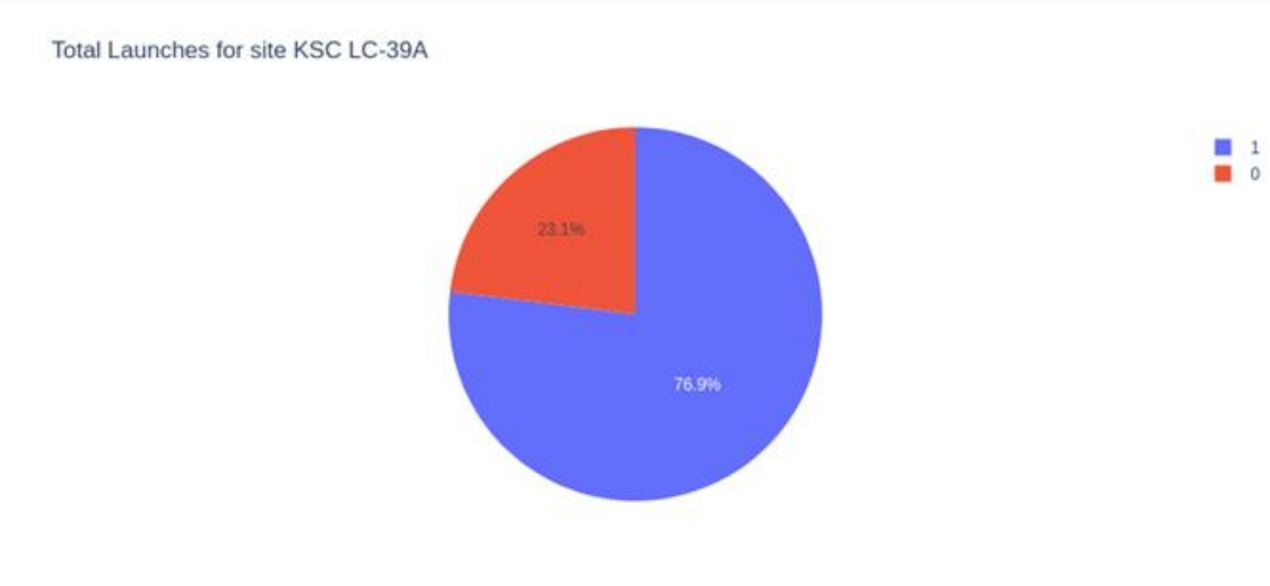
Success Rates

•The **launch site** plays a **critical role in mission success**, as it directly impacts factors such as safety, accessibility, and proximity to key infrastructure.
•Locations with optimal logistical support and safety measures contribute to **higher mission success rates**.



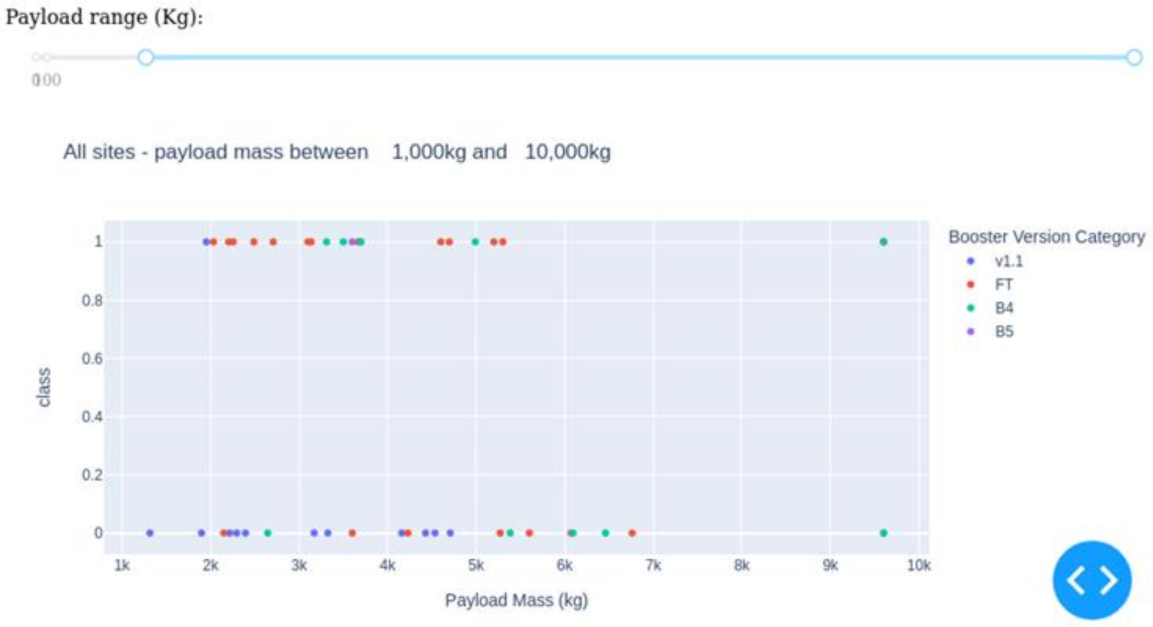
Highest Success Lauched Site

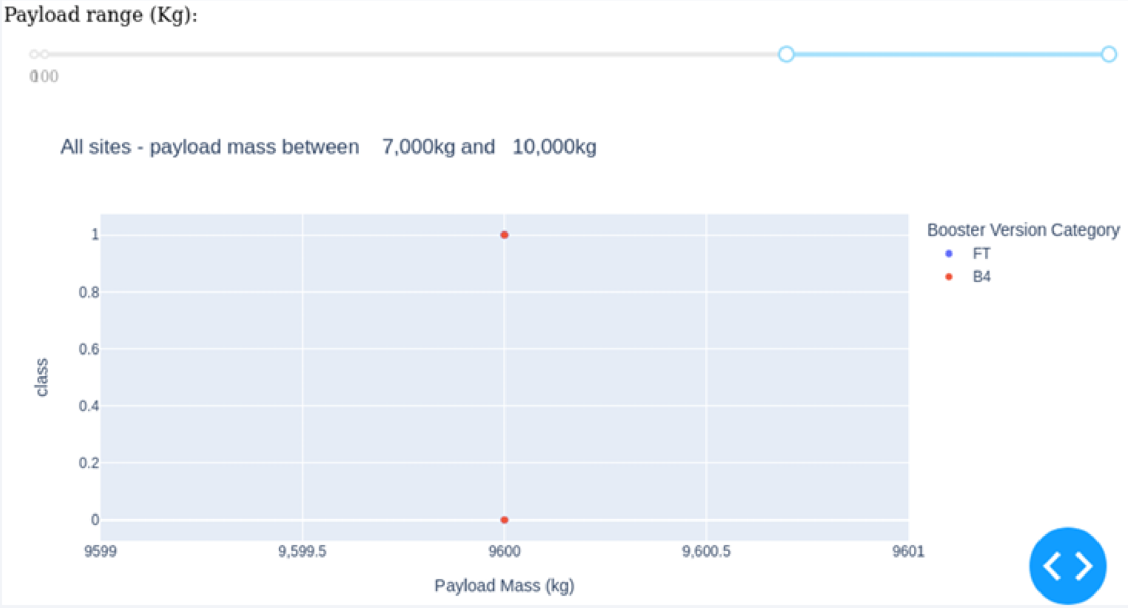
•**76.9% of launches** at this site have been **successful**, indicating a relatively high success rate compared to other locations.
•This success rate highlights the site’s **reliability** and its importance in SpaceX's mission planning.



Payload Mass vs Class

Payloads under 6,000 kg combined with **FT boosters** represent the **most successful combination** for SpaceX launches, showing consistently high success rates.





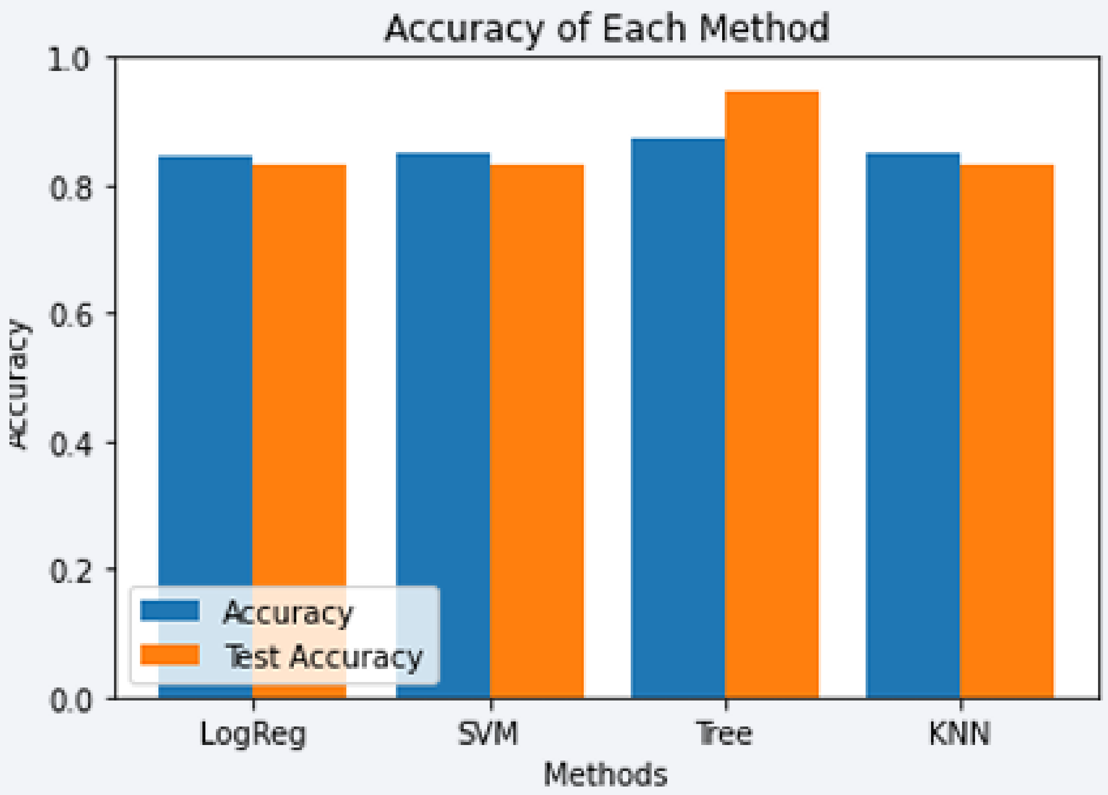
There is insufficient data to accurately assess the risk associated with launches carrying payloads over **7,000 kg**.

Section 5

Predictive Analysis (Classification)

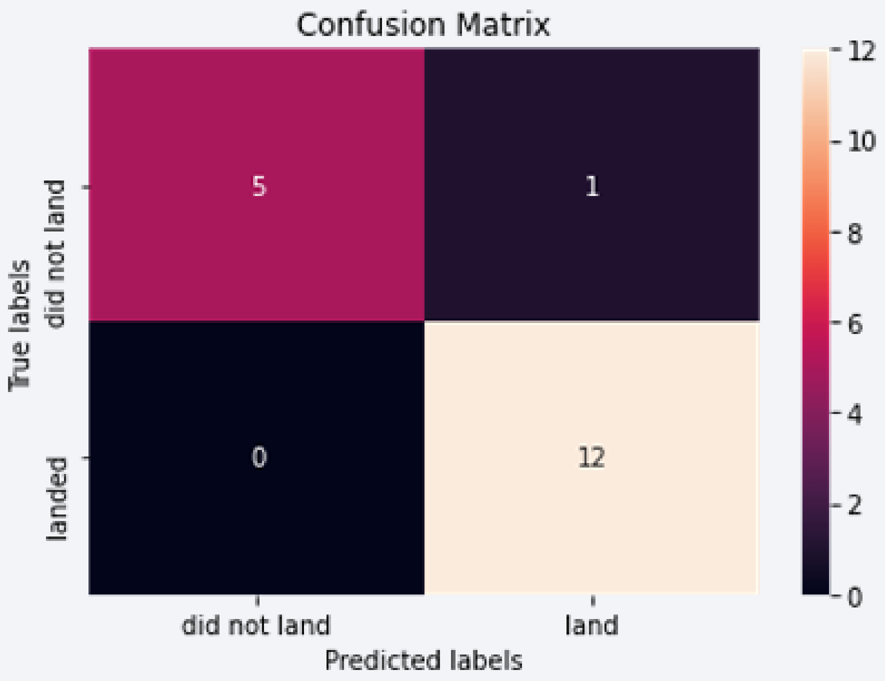
Classification Accuracy

•**Four classification models** were evaluated, and their **accuracies** are shown in the plot beside.
•The model with the **highest classification accuracy** is the **Decision Tree Classifier**, which achieved an accuracy of over **87%**.



Confusion Matrix

The **confusion matrix** for the **Decision Tree Classifier** confirms its high accuracy, displaying a significantly higher number of **true positives** and **true negatives** compared to **false positives** and **false negatives**.



Conclusions

•**KSC LC-39A** stands out as the **best launch site**, showing the highest success rate and offering strong logistical support.
•**Payloads under 6,000 kg** combined with **FT boosters** are the most successful combination, demonstrating consistently high mission success rates.
•The **Decision Tree Classifier**, with an accuracy of over 87%, proves to be an effective tool for predicting **successful landings**, optimizing mission planning.
•**Various data sources** were analyzed, leading to refined conclusions and a deeper understanding of mission outcomes, highlighting the importance of data quality in decision-making.

Appendix

- **Link to the GitHub repository** containing all the code, data, and documents related to the project.

Thank you!